Prediction Is Not Fact

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach predictions as checkable, fallible outputs.

Separate the two rules and quantities, audit observation quality and reserve the Dry1 records for independent evidence.

Keep one habit: Look for linked trial IDs, retained predictions and an explicit assessed denominator.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Separate a recorded **prediction** from the **observation** used to check it

Teach comparison as an **evidence** procedure. The ramp and drying records are fictional; do not present either predictor as a scientific law.

Distinguish predictor **input**s from outcome measurements, correct rule execution from outcome agreement, and an **unassessed** case from a failure. Keep **observation** quality open to documented review.

By the end, I can keep predicted and observed outcomes separate and audit a disagreement.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Ramp trial cards A through F with heights and separate distance cards from the lesson
- A record sheet with trial ID, model version, **input**, **prediction**, **observation**, outcome category and status
- Drying records P through S from W4; paper only and no personal data

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a specific source of uncertainty and explain how it affects a **prediction**

Reference example, not the only acceptable wording: The rule and two stated possibilities are known; the actual height is unresolved. At 5 predict short, at 6 predict tall. Inspect the original readable record or remeasure.

Repeating a calculation supplies no new information about which height was recorded.

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Construct a classifier with a stated **input** format, selected rule and recorded **output**

Reference example, not the only acceptable wording: Require a present height, unit cm, a whole number and range 1 through 10 inclusive.

For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range. Fractional values such as 4.5 cm also fail.

The contract defines accepted **input**s; it does not list only the **fitting** records or establish **prediction** correctness.

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a loop by testing its condition before every repetition

Reference example, not the only acceptable wording: Tests: 2 true, 4 true, 6 true, 8 false. Three additions are three iterations. The loop reports 8; the fourth test does not run the body.

It remains **unassessed**. Obtain a reliable distance record for the same trial, with units and the rule defining far. Repeating the **prediction** does not provide the missing **observation**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Boundaries and 'Not Sure'

Build a Human Classifier

Repeating Efficiently

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

Concept to explain

A **prediction** is a fallible **output** supported by clues.
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Term	Meaning
prediction	estimate about an unknown outcome
evidence	information that supports or challenges a claim
fact	claim established by direct or reliable checking
ground truth	reference outcome used to evaluate a prediction

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach prediction checking with two distinct quantities

> Keep the model rule, observation policy, trial ID and original prediction visible.

Begin with Ramp1’s evidence protocol

A **prediction** states what a model **output**s. An **observation** records what was measured or happened. To check a **prediction**, keep both records and compare the same case under the same **outcome rule**.

Writing a **prediction** into the **observation** column would make the check circular.

Use a fictional ramp investigation. Before each trial, Ramp1 predicts far for a ramp height of at least 6 cm and near otherwise. Its allowed heights are whole centimetres from 1 through 10.

After a toy rolls, a learner measures its travel distance. The **outcome rule** is far for a distance of at least 4 floor units and near otherwise. Ramp height and travel distance are different quantities.

The values in this lesson are invented practice records, not a physical law about ramps. Ramp1 is a fixed paper model for learning how to check **prediction**s.

Each trial has an ID so that its height, **prediction** and later measurement stay together. Keep the model version and record the **prediction** before opening the **observation** card.

Compare established matches and the original mismatch

For B, the **input** 6 cm produces the **prediction** far. The later distance of 3 units produces the observed category near. This is a **mismatch**. Retain far as the original **prediction** and near as the observed category.

Changing the old **prediction** to near would hide the mistake rather than improve the original result.

For A and C, the categories match. These two matches provide **evidence** about these trials. They do not show that every future **prediction** will be correct.

Even a correctly computed **prediction** can disagree with an **observation** because following the model rule and matching the world are different checks.

Model the separate drying trial

A fictional drying trial uses a fixed rule and later measures what actually happened.

Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

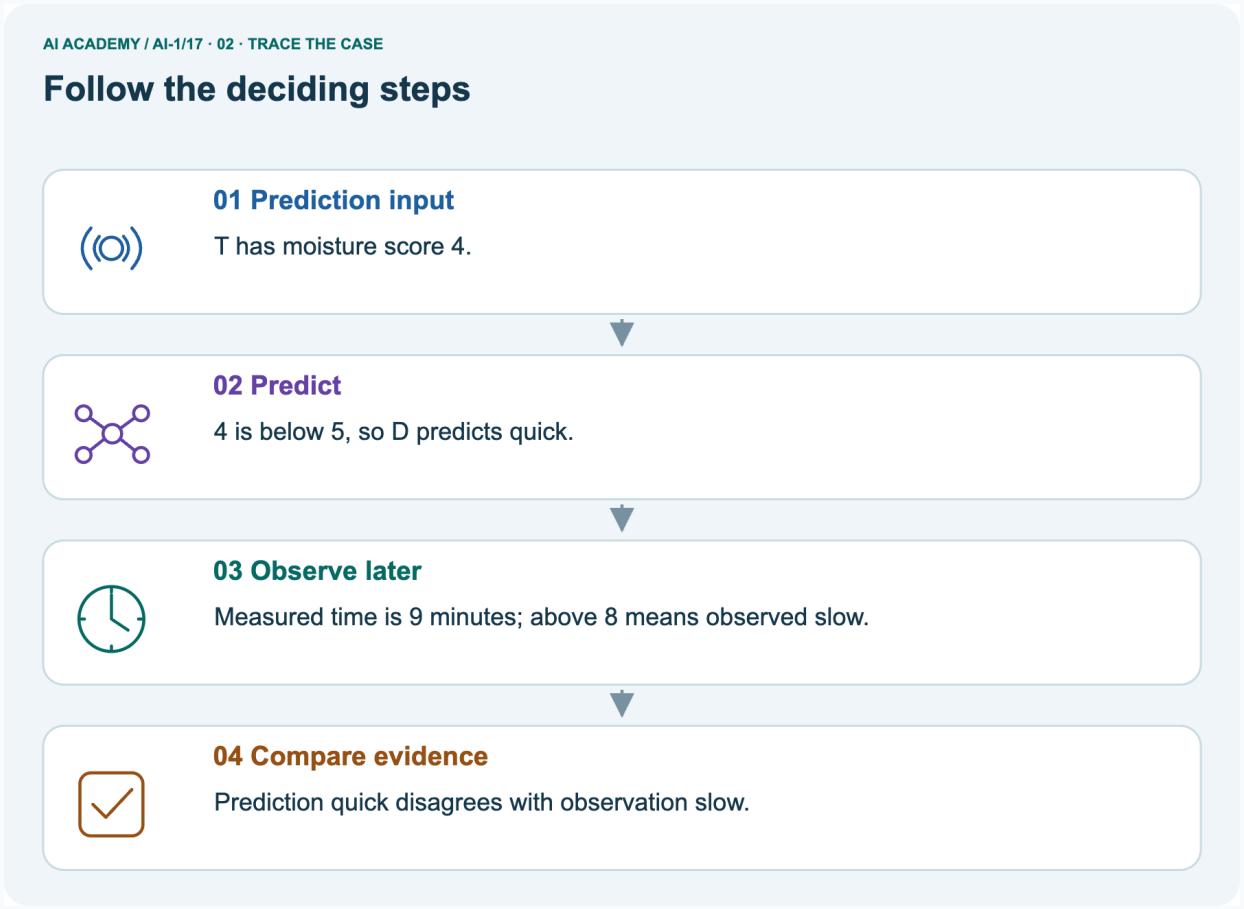
AI ACADEMY / AI-1/17 · 03 · INSPECT THE LESSON MODEL

Two records meet at a comparison

Before the trial	After the trial	Comparison
A: height 4 cm	Distance 3 units	Near matches near
B: height 6 cm	Distance 3 units	Far differs from near
C: height 8 cm	Distance 5 units	Far matches far

Ramp1 predicts from height: A near, B far, C far. The observation is travel distance. Apply the distance outcome rule separately, then compare categories for the same trial.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

Altering the recorded time to match D would destroy the evaluation rather than repair the model.

Change only its later observed time

1. Change: Keep score 4 but measure exactly 8 minutes.
2. **Prediction**: The observed label becomes quick and agrees with the **prediction**.
3. Reason: Above 8 is strict: equality is quick under this policy.

Limit a single trial's result

One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

Handle missing, conflicting and corrected observations

Record	Evidence available	Status under the outcome rule
D: height 5 cm; predicts near	Distance missing	Unassessed; request the trial record
E: height 7 cm; predicts far	Two records say 3 and 4 units	Unassessed; investigate the conflict
F: height 6 cm; predicts far	Verified distance exactly 4 units	Match; at least 4 includes 4

For E, 3 units means near and 4 units means far. The disagreement can change the judgment. Check the trial ID, original measurement and recording procedure. Do not choose the distance that makes Ramp1 look better.

If the conflict cannot be resolved, retain both reports and mark the comparison **unassessed**.

An **observation** is **evidence**, not an automatic guarantee of truth. A wrong trial ID, unclear unit or copying error can make the comparison unreliable.

If a supported correction is made, retain the original entry, the correction, its source and the reason. Remeasuring a new trial does not reveal what happened in the old one.

Use the picture deliberately

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT** available **evidence**

> **2. Process — the deciding step**
>
> **PROCESS** use a rule or model to estimate an unseen outcome

> **3. Output — the result**
>
> **OUTPUT prediction** that must remain separate from verified **fact**

CORE IDEA A **prediction** is a fallible **output** supported by clues.

UNDER THE HOOD **Prediction** estimates a target that is unknown at decision time. **Evidence** supports the estimate; **ground truth** is the later or independently established outcome.

Treating **prediction**s as labels can create self-confirming records.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A rain **prediction** uses weather clues, but the later **observation** of rain or no rain is the **fact** that evaluates it.

STEP 1 — OBSERVE available **evidence**

STEP 2 — TRACE use a rule or model to estimate an unseen outcome

STEP 3 — CONCLUDE **prediction** that must remain separate from verified **fact**

MODEL REASONING Mark model or rule **output**s as **prediction**s, observable clues as **evidence**, and independently checked outcomes as **fact**s.

A **prediction** becomes evaluated against a **fact**; it does not become a **fact** merely because time passes.

DO NOT OVERCLAIM Do not copy a model **prediction** into the answer column and later call it **ground truth**.

What new **evidence** could change this answer?

Concrete teaching case

A weather helper predicts tomorrow's rain.

Is this a **fact** about tomorrow? How can it be checked?

Exact explanation to model

It is a **prediction**. Record it now and compare with a predefined rain **observation** tomorrow. Useful **prediction**s are not guarantees.

Explain the added worked example

Key terms

Term	Meaning
Prediction	A model output recorded before consulting the comparison outcome
Observation	A recorded measurement or event used as evidence
Outcome rule	The agreed method for turning observed evidence into the category being checked
Mismatch	A prediction that differs from an established comparison outcome
Unassessed	A case whose prediction has not yet been judged against adequate outcome evidence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

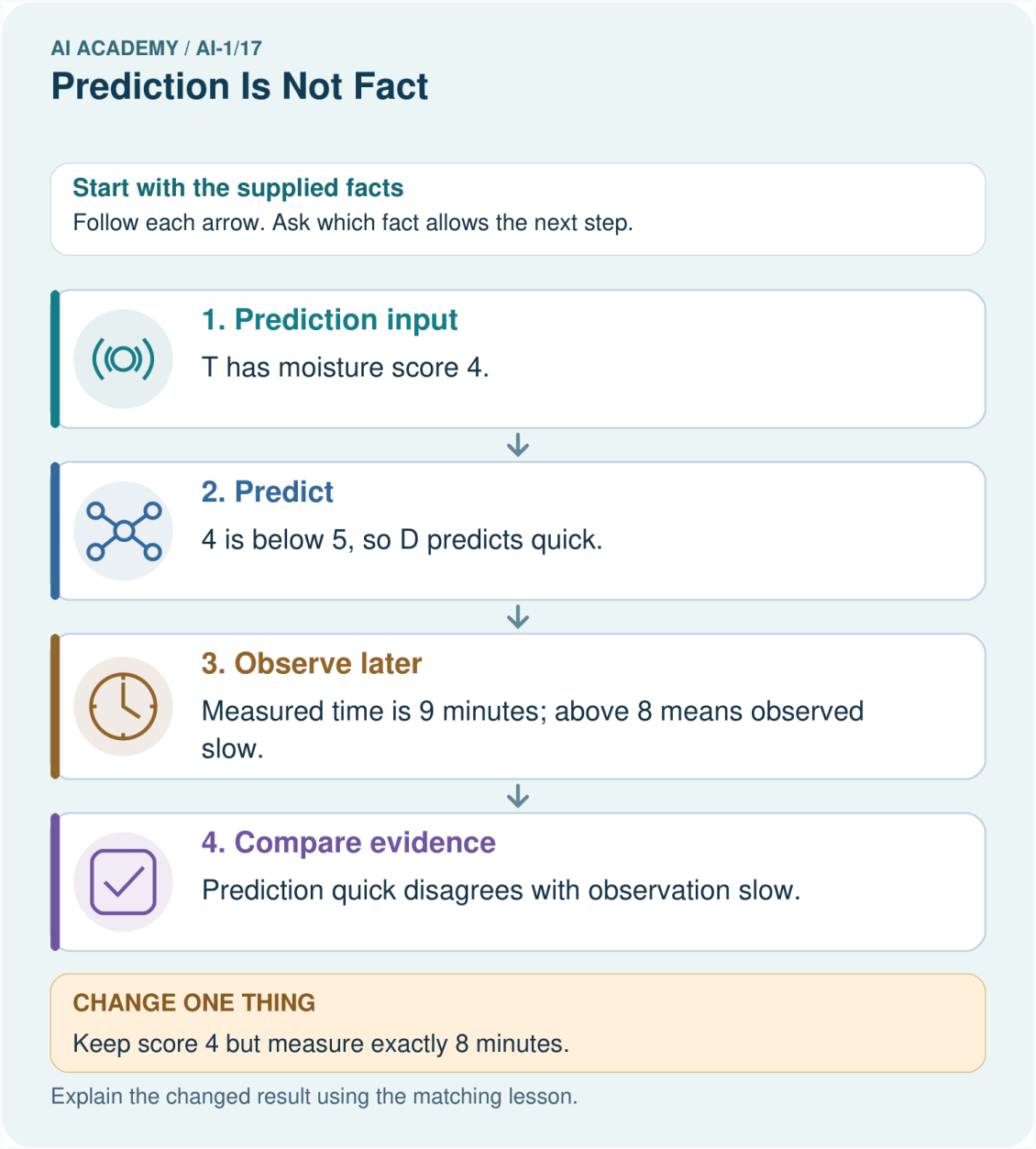
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Prediction Is Not Fact

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Prediction input	T has moisture score 4.
2. Predict	4 is below 5, so D predicts quick.
3. Observe later	Measured time is 9 minutes; above 8 means observed slow.
4. Compare evidence	Prediction quick disagrees with observation slow.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the **outcome rule**. A partner says exactly 4 is near because far means greater than 4.

Use the stated rule to resolve the disagreement.

Expected reasoning and fading help

Teacher check: Ramp1 predicts far at height 6. The **observation** is far because the stated **outcome rule** is at least 4, including equality. The **prediction** matches.

Greater than 4 would be a different rule and must not replace the agreed one after viewing the result.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Prediction**s and **observation**s. Proposed explanation: “Running the rule correctly proves the **prediction** correct”. Decide whether this explanation is supported.

Point to the step or supplied **fact** that decides; explain your repair if needed.

Expected correction: Rule execution and agreement with the observed outcome are separate checks.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Which field changes the **observation** label, and which field changes the model **prediction**?

Time changes the **observation**; moisture score changes the **prediction**. At score 4 and time 8, both labels are quick.

Changed-case reasoning: The observed label becomes quick and agrees with the **prediction**. Above 8 is strict: equality is quick under this policy.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which **fact**s are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given **fact**s: A weather helper predicts tomorrow's rain.

Write the question to explain: Is this a **fact** about tomorrow? How can it be checked?

Reveal after **prediction**s: It is a **prediction**. Record it now and compare with a predefined rain **observation** tomorrow. Useful **prediction**s are not guarantees.

Use the earlier changed question as optional later practice. Reserve Transfer the check to drying **prediction**s for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

A predicts near, observed near, match; B far, observed near, **mismatch**; C far, observed far, match. D predicts near with missing distance; E predicts far with conflicting 3 and 4; both **unassessed**.

Two matches out of three assessed, two **unassessed**. Exit: changing B after reading the outcome hides a **mismatch**; retain its original far **prediction** and observed near outcome, and identify any revised model separately.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve why a missing observation prevents judgment even when a prediction is available.
7 to 20 minutes	Model A, B and C with separate height and distance cards. Record predictions before revealing distances. Explain that the lesson shows a worked retrospective example.
20 to 33 minutes	Complete W1 and W2. Require both numerical rules and include the exact-distance boundary at 4.
33 to 45 minutes	Discuss D and E and complete W3. Ask which evidence could establish the old trial outcome and why a new trial cannot do so.
Session 2 start 0 to 8 minutes	Retrieve input, prediction, observation and outcome category with four labelled spaces.
8 to 23 minutes	Complete the guided A-through-E results record. Make assessed and unassessed counts visible before forming a result fraction.
23 to 38 minutes	Collect W4 independently. Watch for the different greater-than boundary at 10 minutes and the missing S outcome.
38 to 45 minutes	Discuss W5 and the exit. Preserve old predictions, distinguish documented observation corrections, and connect model revision to new reserved tests.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

CartC predicts NEAR,FAR,FAR,FAR for A,B,C,D. The supplied distances establish NEAR for A and B: A matches, B mismatches. C is unassessed because its distance is missing. D is unassessed because 4 and 6 fall on different sides of the outcome boundary. Report one match out of two assessed trials, with two unassessed, and keep the result limited to this evidence. Reserve Transfer the check to drying predictions for the individual first attempt.

Run and record

1. Complete the task: Separate **prediction**, **evidence**, and **fact** in four scenarios.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

Running the rule correctly proves the **prediction** correct: Rule execution and agreement with the observed outcome are separate checks.

A missing outcome is a success or a failure: Mark it **unassessed** and report the missing count alongside assessed results.

Every recorded **observation** must be true: Check the trial identity, units and **evidence**; document supported corrections.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Transfer the check to drying predictions for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR A **prediction** is a **fact** because a computer produced it.

For guessing, return to given **fact**s and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A new trial repairs the old **observation**: A new trial has its own ID and result; the old outcome may remain unknown.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

P predicts quick, observed slow, **mismatch**. Q predicts slow, observed quick because exactly 10 is not greater than 10, **mismatch**. R predicts slow, observed slow, match. S predicts quick, observed outcome unavailable, **unassessed**.

One match out of three assessed cases, with one **unassessed**. This describes the assessed cases and does not establish the result for S or future trials.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Separate inputs from observed outcomes

For Ramp1 trial B, height is 6 cm and the later travel distance is 3 units. State the **input**, **prediction**, **observation**, observed category and comparison status.

Explain why the **prediction** rule and **outcome rule** use different measurements.

Input height 6 cm; **prediction** far because 6 is at least 6; observed distance 3 units; observed category near because 3 is below 4; **mismatch**.

Height is available to the model before the trial, while travel distance is measured afterward to check its **prediction**.

W2 Check a boundary without changing the rule

Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the **outcome rule**. A partner says exactly 4 is near because far means greater than 4. Use the stated rule to resolve the disagreement.

Ramp1 predicts far at height 6. The **observation** is far because the stated **outcome rule** is at least 4, including equality. The **prediction** matches.

Greater than 4 would be a different rule and must not replace the agreed one after viewing the result.

W3 Keep incomplete evidence visible

D predicts near but has no distance record. E predicts far but its two distance records say 3 and 4 units.

State each comparison status, request useful **evidence**, and explain why neither guessing nor a new trial can establish the old trial's outcome.

Both are **unassessed**. For D request the original distance record for D with units. For E inspect the trial IDs, original measurement and recording procedure to resolve 3 versus 4.

Guesses supply no **observation**; a new trial produces a new outcome. Retain unresolved **evidence** if the old result cannot be established.

W5 Correct a report without erasing evidence

After seeing B's observed near category, a learner replaces Ramp1's far **prediction** with near. They call the new entry a corrected **observation** and report a perfect result for A through C.

Explain the two record-keeping errors and write a defensible replacement report.

They changed a **prediction** after seeing the outcome and mislabelled that change as an **observation** correction. Restore the original far **prediction** for B and retain the observed near category.

Report two matches out of three assessed Ramp1 trials A through C, including B's **mismatch**. Any revised model needs a new version and new reserved cases for evaluation.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

It is a **prediction**. Record it now and compare with a predefined rain **observation** tomorrow. Useful **prediction**s are not guarantees.

Additional transfer practice question

A model predicts rain tomorrow. Tomorrow is dry. Was the **prediction** an **observation** of tomorrow? Explain what new **evidence** settles the result.

Reasoned practice answer

No. A **prediction** estimates an unknown outcome. Tomorrow's actual weather settles whether it was correct. A plausible **prediction** can still be wrong.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **prediction** is a **fact** because a computer produced it.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

For example, record precipitation observed at the school weather station between 00:00 and 24:00 Tuesday above its stated detection threshold. Damp ground alone has another possible cause.

Save **prediction** time, location and **observation** method before evaluating the result.

Explanations for the additional workbook examples

W1 Separate inputs from observed outcomes

W2 Check a boundary without changing the rule

W3 Keep incomplete evidence visible

W4 Transfer the check to drying predictions

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

P predicts quick, observed slow, **mismatch**. Q predicts slow, observed quick because exactly 10 is not greater than 10, **mismatch**. R predicts slow, observed slow, match. S predicts quick, observed outcome unavailable, **unassessed**.

One match out of three assessed cases, with one **unassessed**. This describes the assessed cases and does not establish the result for S or future trials.

W5 Correct a report without erasing evidence

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Label the sorter's suggested bin as a **prediction** until a human confirms the final placement.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Use separate coloured cards for **prediction**s and **observation**s and pair them by trial ID. Read at least and greater than aloud on number lines.

Extension: investigate how selective missing records could make the assessed cases an incomplete picture of model performance.

W4 is secure with quick/slow/slow/quick **prediction**s, slow/quick/slow/unknown outcomes, two **mismatch**es, one match and one **unassessed** case. Require one of three assessed and limited claims.

After support retry scores 5, 8, 4 with times 9, 10, missing: quick/slow/quick **prediction**s, quick/quick/unknown outcomes; one match, one **mismatch**, one **unassessed**.

Plan the delayed check

Suggested return points in this level: ai-1/18, ai-1/20, ai-1/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 18: Confidence Is Not Correctness. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 18 distinguishes confidence from correctness and asks what **evidence** could support a confidence claim.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Separate a recorded **prediction** from the **observation** used to check it

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Trial F has height 6 cm and verified travel distance exactly 4 units. Apply Ramp1 and the **outcome rule**. A partner says exactly 4 is near because far means greater than 4.

Use the stated rule to resolve the disagreement.

Reasoned practice answer: Ramp1 predicts far at height 6. The **observation** is far because the stated **outcome rule** is at least 4, including equality. The **prediction** matches.

Greater than 4 would be a different rule and must not replace the agreed one after viewing the result.

Check every learner before discussion: Use the worked case for **Prediction**s and **observation**s. Proposed explanation: "Running the rule correctly proves the **prediction** correct". Decide whether this explanation is supported.

Point to the step or supplied **fact** that decides; explain your repair if needed.

Misconception repair: Rule execution and agreement with the observed outcome are separate checks.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Keep score 4 but measure exactly 8 minutes. Require a **prediction** and a reason before comparing. Expected result: The observed label becomes quick and agrees with the **prediction**.

Reason: Above 8 is strict: equality is quick under this policy.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

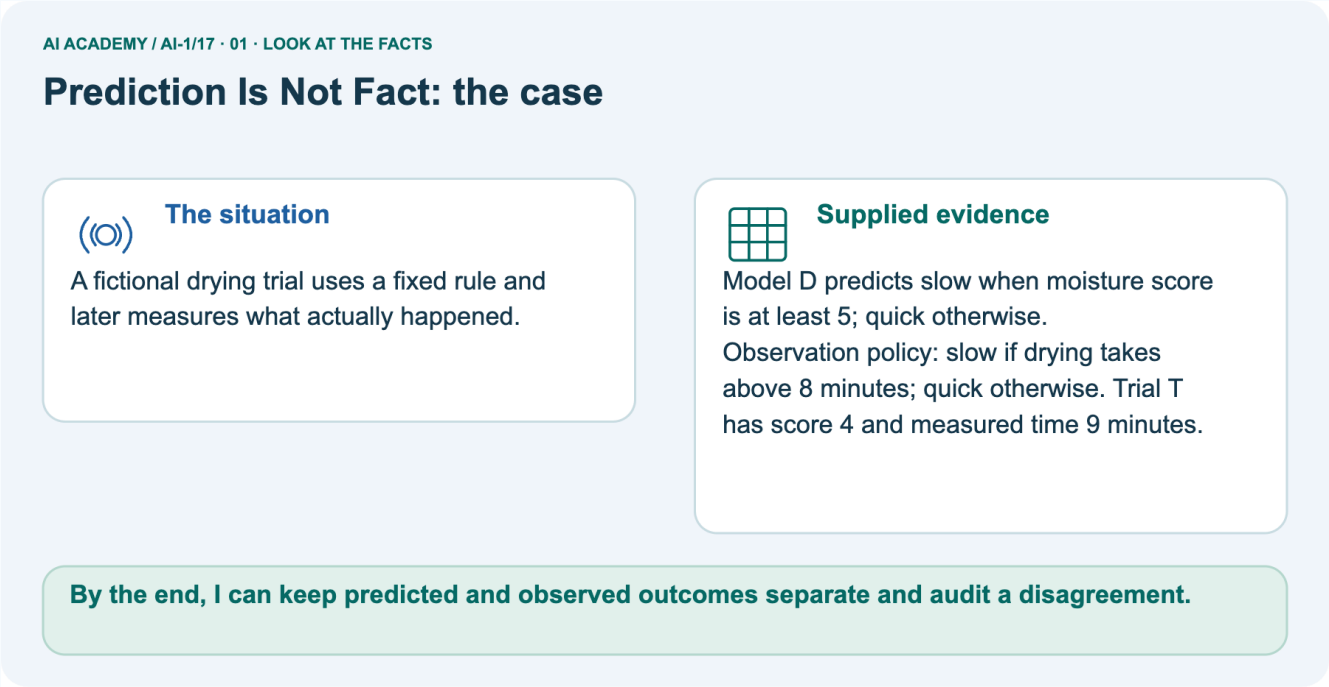
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Prediction Is Not Fact: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

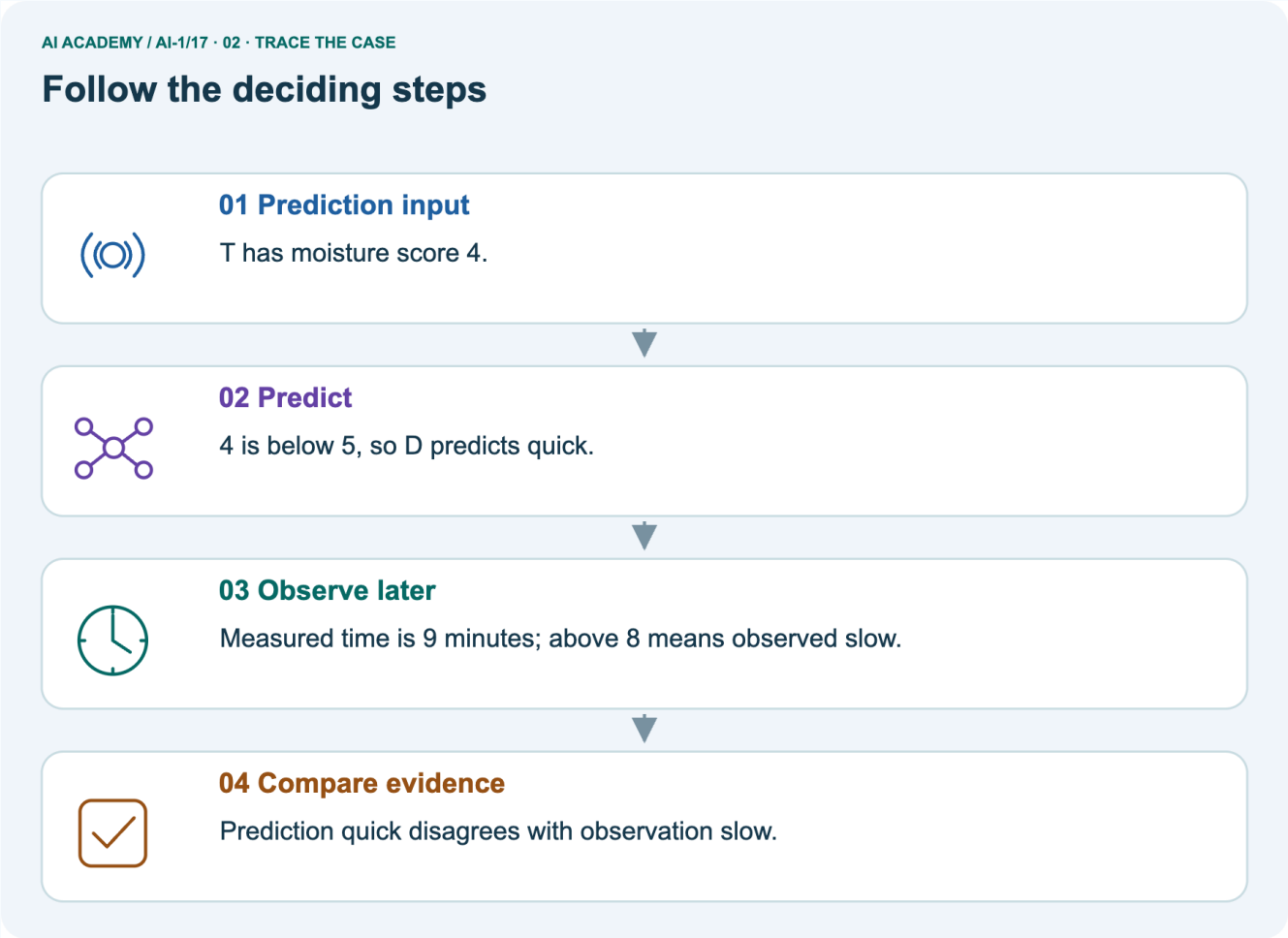
A fictional drying trial uses a fixed rule and later measures what actually happened. Supplied **fact**s: Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

Point to the given **fact** that matters most. Explain why it matters.

Teacher prompt: Point to the given **fact** that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



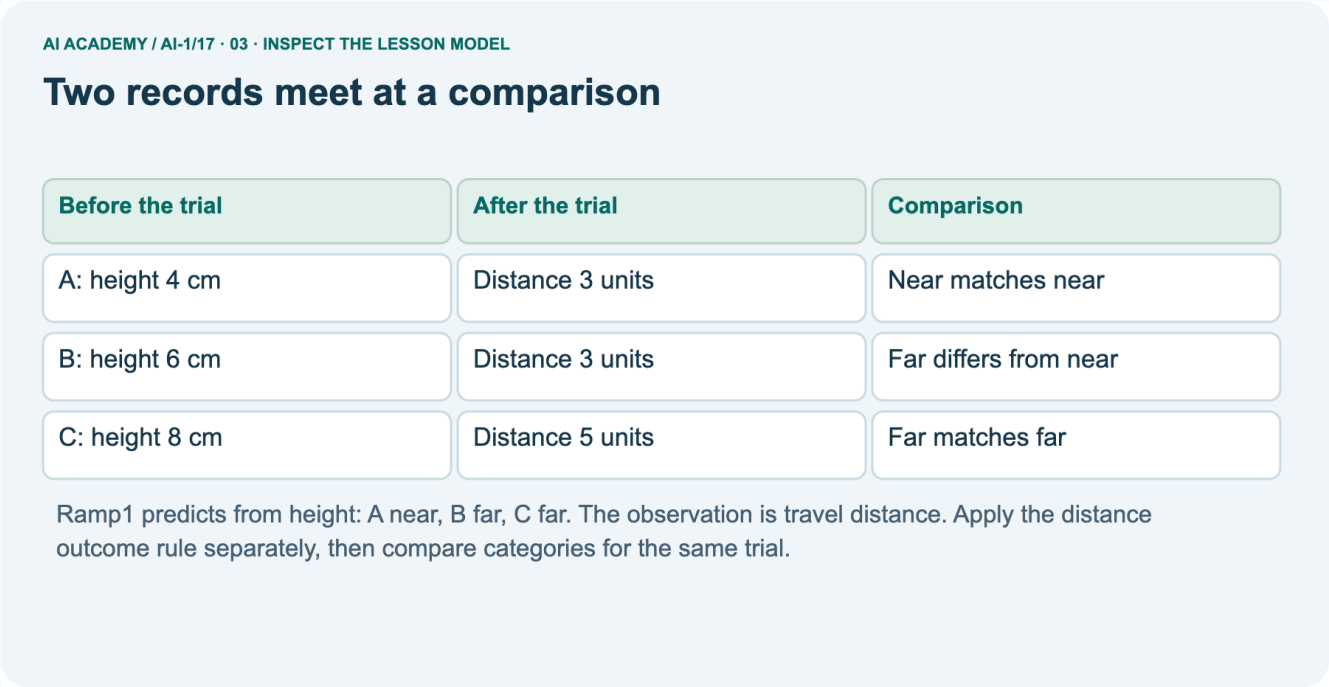
Follow the labelled picture with the matching explanation. Open the vector diagram

Prediction input: T has moisture score 4. → Predict: 4 is below 5, so D predicts quick. → Observe later: Measured time is 9 minutes; above 8 means observed slow. → Compare **evidence**: **Prediction** quick disagrees with **observation** slow.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two records meet at a comparison



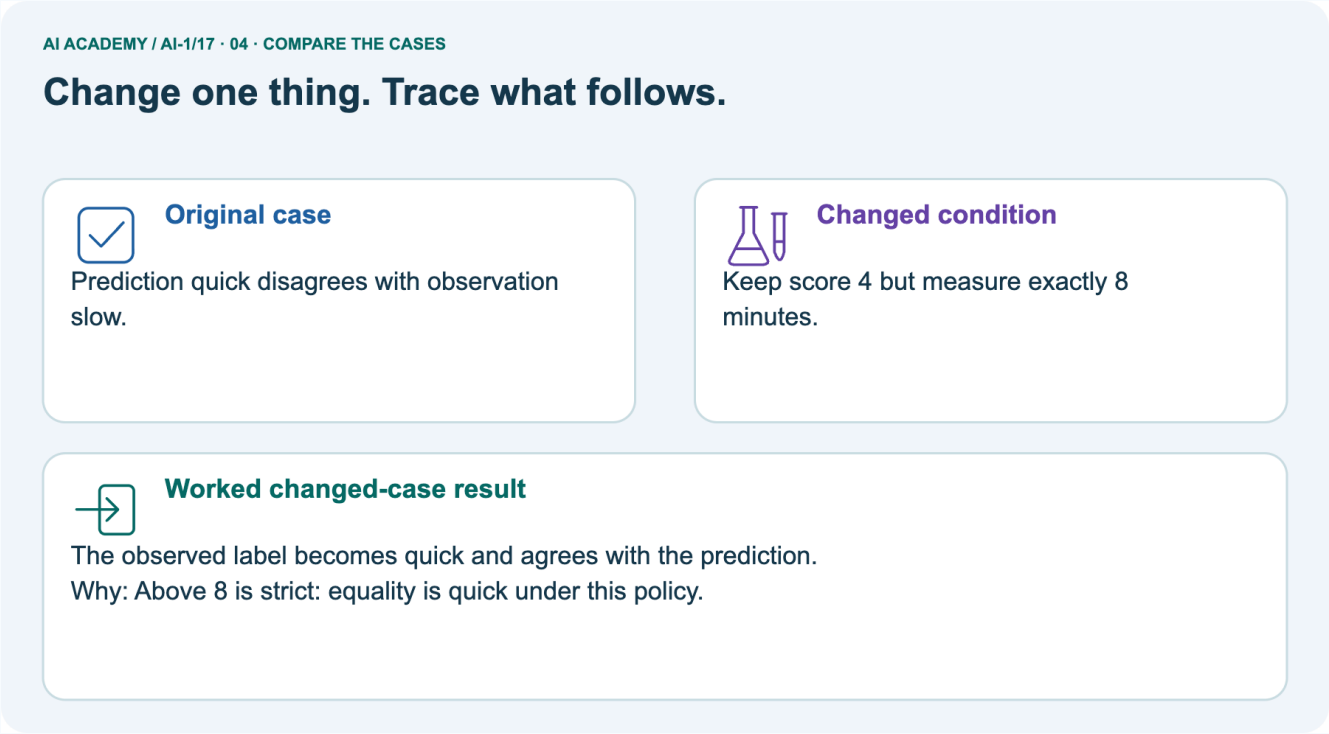
Follow the labelled picture with the matching explanation. Open the vector diagram

Two records meet at a comparison. A: height 4 cm | Distance 3 units | Near matches near. B: height 6 cm | Distance 3 units | Far differs from near. C: height 8 cm | Distance 5 units | Far matches far

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Prediction** quick disagrees with **observation** slow. Changed condition: Keep score 4 but measure exactly 8 minutes. Expected result: The observed label becomes quick and agrees with the **prediction**.

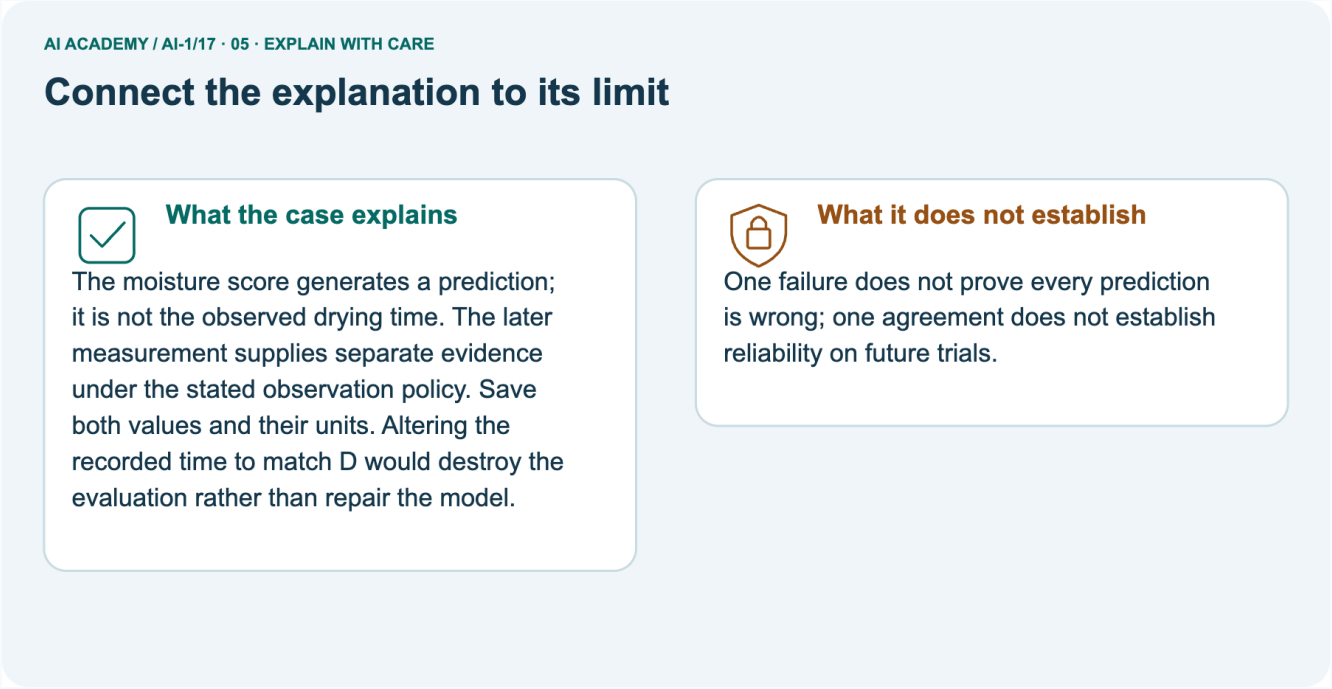
Reason: Above 8 is strict: equality is quick under this policy.

Predict before checking the worked changed-case result. Name what changed and what the supplied **fact**s allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied **fact**s allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Above 8 is strict: equality is quick under this policy.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

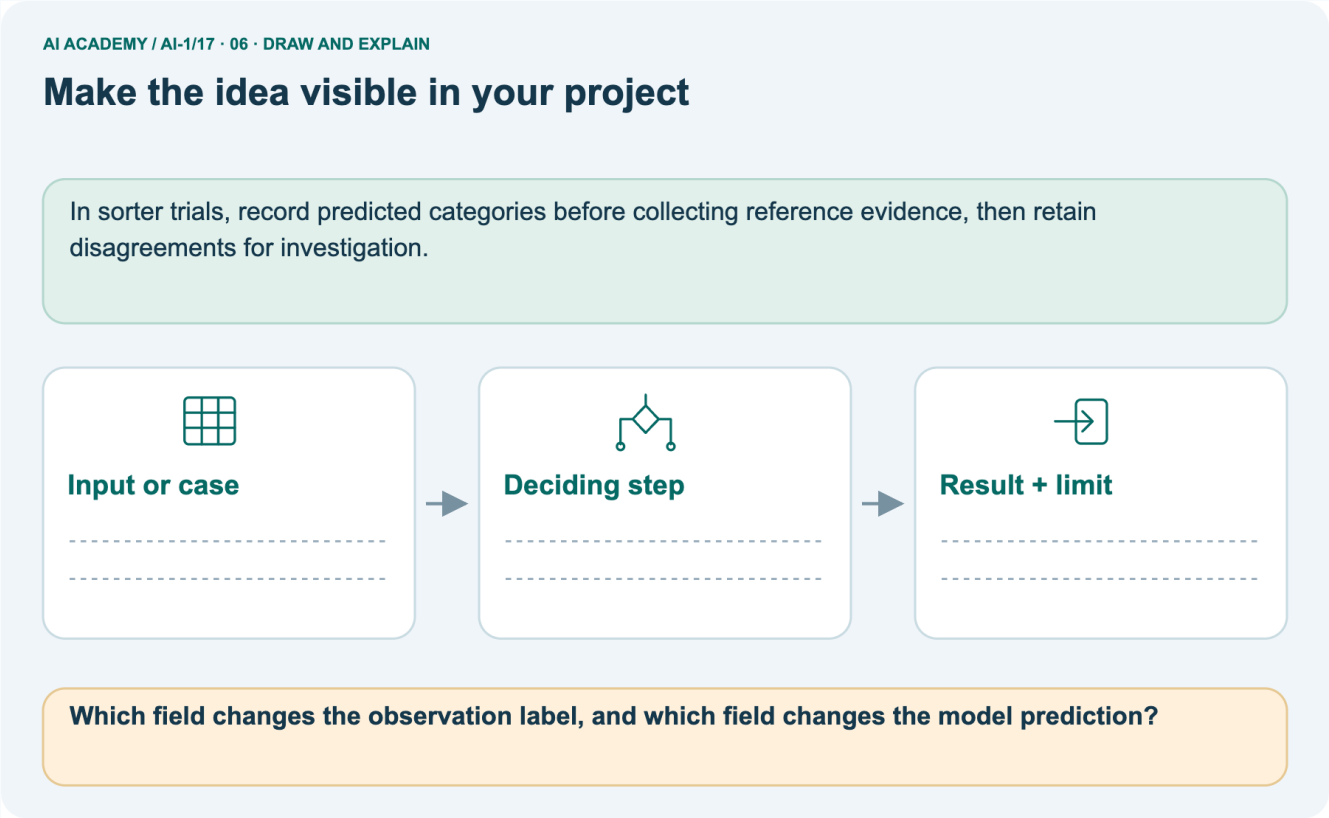
Altering the recorded time to match D would destroy the evaluation rather than repair the model. Limit: One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation. Draw your **input**, deciding step, result and limit.

Supported question: Which field changes the **observation** label, and which field changes the model **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Time changes the **observation**; moisture score changes the **prediction**. At score 4 and time 8, both labels are quick.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Prediction Is Not Fact: the case

02. Follow the deciding steps

03. Two records meet at a comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.